

Management of penetrating neck injuries at a London trauma centre

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Abstract Penetrating neck injuries (PNIs) are uncommon in the UK. The majority of guidelines are from the USA or South Africa. No UK national guidelines exist. Increasing urban violence in the UK has led to an increase in PNIs. There is a need to develop a PNI guideline that reflects the pathology and experience in the UK. A retrospective review of all PNIs managed at St George's Hospital over an 18 month period was undertaken. Data collected included patient demographics, mechanism of injury, investigations, multidisciplinary team involvement and management. Clinical activity data was correlated to current worldwide literature and a flow-chart style clinical guideline was produced. 25 neck wounds were managed over an 18 month period. 68 % were male, 32 % female with a mean age of 36 years. The mechanism of injury included deliberate self-harm (48 %), stab wounds (32 %), gunshot wounds (4 %), shotgun wounds (4 %) and other accidental causes (12 %). 52 % of wounds were superficial to platysma. 58 % of deep wounds had CT. 42 % of patients with deep wounds also had panendoscopy. Interventional radiology was used in one case (8 %) and a single case was managed jointly with the vascular team (8 %). UK ENT surgeons have limited exposure to neck trauma and dedicated head and neck out-of-hours cover is uncommon. There is a need for UK PNI guidelines that reflect local pathology and experience. The St George's PNI guideline can be used to facilitate assessment, documentation and management of a relatively infrequent emergency presentation.

Keywords Neck injuries · Trauma · Neck · ENT · Penetrating wounds

Introduction

Penetrating neck injuries (PNIs) are rare in the United Kingdom. However, increasing urban violence has led to a rising incidence [1, 2]. The management of these injuries remains controversial with most experience and the majority of guidelines originating from North America and South Africa. Conversely, there is a relative paucity of data specific to penetrating neck trauma in the United Kingdom where the social demographics and patterns of injury are different. St George's Hospital has recently become a major trauma centre and thus there is increased need to develop a PNI guideline reflecting the pathology and experience of such injuries in the UK. A PNI management algorithm is herein produced, based on the St George's Hospital experience and a review of the worldwide literature Table 1.

The management of PNIs has evolved significantly over the last century. Trauma surgery has seen much advancement during military conflict. Mortality rates from PNIs were 15 % in the American Civil War [3] and 11 % in World War I [4], during which there was a tendency towards expectant management which often resulted in uncontrollable haemorrhage, the largest cause of death in patients sustaining these injuries. Towards the end of World War II, there was a shift from expectant management to mandatory surgical exploration for all patients with penetrating neck injuries, with one series showing a mortality rate of 7 % in patients sustaining PNIs [5]. As a result, routine exploration of penetrating neck trauma was strongly advised [5, 6]. In 1956, Fogelman and Stewart concluded

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Table 1 Zones, mechanisms, descriptions, investigations and management of PNIs at St George's Hospital

Case	Zone	Mechanism	Nerve/Airway/ Vascular Damage	Investigations	Management
1	I	Stab wound	None	FNE, CT, Panendoscopy	Delayed exploration
2	I	Accident	Greater auricular nerve	CT	Delayed exploration
3	II	Stab wound	None	Duplex ultrasonography	Delayed exploration
4	II	DSH	None	FNE	Delayed exploration
5	II	DSH	CNX, IJV	No investigations	Immediate exploration
6	II	DSH	AJV	No investigations	Immediate exploration
7	II	Stab wound	AJV	FNE, CT	Delayed exploration
8	II	DSH	None	FNE, Panendoscopy	Delayed exploration
9	II	DSH	EJV, AJV	CT, FNE, Panendoscopy	Delayed exploration
10	II	DSH	Trachea	Panendoscopy, CT	Tracheostomy, Delayed exploration
11	I, II	Stab wound	CNVII, ECA	Panendoscopy, CT, FNE	Immediate exploration
12	I, II, III	50 wounds from shotgun shell	Vertebral artery	CT, CTA	Vascular stenting Delayed exploration

DSH deliberate self-harm *CNX* vagus nerve *CNVII* facial nerve *AJV* anterior jugular vein *IJV* internal jugular vein *EJV* external jugular vein *FNE* fiberoptic nasendoscopy

that early mandatory surgical exploration of civilian PNIs was the best approach, and this became the classic management of penetrating injuries to the neck [7]. This aggressive management philosophy was driven by the fear of vascular injuries missed at physical examination and consequent neurological complications, and rapidly became the mainstay of management at most major trauma centres.

The 1980s saw a paradigm shift from mandatory exploration towards selective management of penetrating neck injuries, as the high rates of negative exploration were deemed unacceptable. An 18 year prospective study by Biffi et al. showed that selective management of PNIs based on anatomical zone of injury as well as clinical findings was safe and cost-effective [8]. These findings have been supported by many other studies [9–15], and most trauma centres have now adopted the approach of selective management based on a combination of clinical examination and radiographic investigation. Over the last decade, advances in imaging have consolidated the role of selective management as the paradigm. In particular, developments in multislice and helical computed tomography angiography (CTA) have provided a fast, accurate non-invasive method of evaluating patients with penetrating neck injuries. Several studies have shown that CTA is as useful as traditional catheter-based angiography in evaluating vascular injuries in neck trauma whilst providing useful information about other structures which may be injured such as the upper aerodigestive tract and cervical spine [16–20]. Furthermore, the use of CTA can drastically reduce rates of negative surgical exploration [20–22].

By definition, PNIs are injuries that penetrate the full thickness of platysma. Penetrating injuries to the neck may be classified into three zones according to location of site of entry as described by Monson et al. [23]: zone I: demarcated by the clavicles and the cricoid cartilage; zone II: extending from the cricoid cartilage to the level of the angle of the mandible; zone III: extending from the angle of the mandible to the base of the skull.

The management of neck injuries is guided by the zones involved, although it should be remembered that these zones only apply to the anterior neck.

Zone II is readily surgically accessible and exploration is favourable: this has resulted in high rates of negative neck explorations, with figures ranging from 50–60 % [24, 25]. Access to Zone I and zone III injuries is more challenging and may involve opening the thoracic or intracranial cavity, thus these injuries were commonly evaluated with angiography, endoscopy, laryngoscopy and oesophagography.

Major trauma relates to serious, often multiple injuries where there is strong potential to cause death or serious disability. It was recently identified that major trauma care in the United Kingdom could be improved by the implementation of major trauma systems, potentially saving an additional 450 to 600 lives per year. Major trauma systems aim to improve patient outcomes by ensuring that appropriate specialist care is available to meet the urgent needs of trauma patients, and that patients are transferred to the most appropriate unit according to the severity of their injuries. Each major trauma centre must provide 24 hour emergency services catering for the management of serious

injuries, including access to emergency radiology and operating theatres [26].

Strategic health authorities manage NHS England locally and are thus responsible for the establishment of trauma networks, including the designation of major trauma centres and local trauma units. London is one of ten strategic health authorities within NHS England; the London Strategic Health Authority has implemented a trauma system which consists of four trauma networks, each based on a major trauma centre working with a collection of local trauma units. St George's Hospital is one of four major trauma centres in London.

Methods

St George's Hospital is the major trauma centre for the South West London and Surrey Trauma Network, providing 24 hour emergency surgical services across all surgical subspecialties as well as interventional radiology services for a population of approximately 1.3 million. All patients presenting to St George's Hospital between December 2009 and May 2011 with neck trauma were identified by a search of the ICD-10-based hospital database.

Results

25 patients with neck wounds were identified. Of these, there were 21 patients: one patient was admitted three times and two patients were admitted twice each over 18 months. 68 % of admissions were male ($n = 17$) and 32 % were female ($n = 8$). The mean ages at admission were 38.4 years (± 22.2) for males and 33.4 years (± 9.8) for females. 48 % of presentations resulted from deliberate self-harm ($n = 12$), 32 % from stab wounds ($n = 8$), 12 % from accidents ($n = 3$), 4 % from gunshot wounds ($n = 1$) and 4 % from shotgun wounds ($n = 1$).

It was found that a complete head and neck examination was documented in 16 % of cases ($n = 4$). 28 % of cases were examined by flexible fiberoptic nasendoscopy ($n = 7$). 48 % had regional nerve function assessed ($n = 12$).

The patients were haemodynamically stable in all 25 cases. 12 cases involved injuries violating platysma: 50 % resulted from deliberate self-harm ($n = 6$), 33.3 % from stab wounds ($n = 4$), 8.3 % from an accident ($n = 1$) and 8.3 % from shotgun wounds ($n = 1$). There were two cases of zone I injury and eight cases of zone II injury. One case involved both zone I and zone II and another involved injuries across all three zones. 50 % of cases had vascular involvement ($n = 6$): in four cases the damaged vessel was a vein and in two cases an artery was damaged; the most commonly injured blood vessel was the anterior jugular

vein, damaged in three cases. Nerve injury occurred in 25 % of cases of PNI ($n = 3$) and the upper aerodigestive tract was damaged in 8.3 % of cases ($n = 1$).

58.3 % of PNIs were investigated by CT ($n = 7$), 8.3 % underwent CTA ($n = 1$) and 8.3 % received assessment for vascular injury by Duplex ultrasonography ($n = 1$). Panendoscopy was performed on 41.7 % of cases ($n = 5$). All 12 cases proceeded to theatre, with two cases undergoing immediate surgical exploration and one case involving interventional radiological input.

We have produced a penetrating neck trauma management algorithm based on the analysis of our figures and the worldwide literature (Fig. 1).

Discussion

Initial assessment

There is little dispute that an unstable patient should undergo immediate surgical exploration [9, 12, 20]. In our series, there were no unstable patients unresponsive to resuscitation.

Initial assessment should follow Advanced Trauma Life Support (ATLS) principles. Flexible fiberoptic nasendoscopy is essential for the examination of the endolarynx in the stable patient: lacerations, haematomas, vocal cord mobility and oedema should be identified at the time of initial assessment [27]. The marginal mandibular, hypoglossal and accessory nerves are particularly vulnerable to penetrating neck trauma. Careful examination and documentation of cranial nerve function is therefore particularly important. Documentation of the fiberoptic assessment and cranial nerve function was found to be poor in the St George's series. The St George's penetrating neck injury guideline acknowledges this by directing the clinician to perform a full head and neck examination including fiberoptic nasendoscopy and cranial nerve assessment where possible.

If the platysma is intact then by definition the neck wound is superficial. This fact is recognised by many international management algorithms and is used to guide treatment [28, 29]. In 48 % of neck wounds presenting to St George's hospital, the platysma was breached. It was found that local anaesthetic facilitated a more accurate assessment of the wound. We recommend that the attending emergency otolaryngology service should be involved in making an assessment in all cases of penetrating neck injuries.

Further investigation

Much of the literature suggests that injuries to zone II of the neck do not require evaluation by CT and that

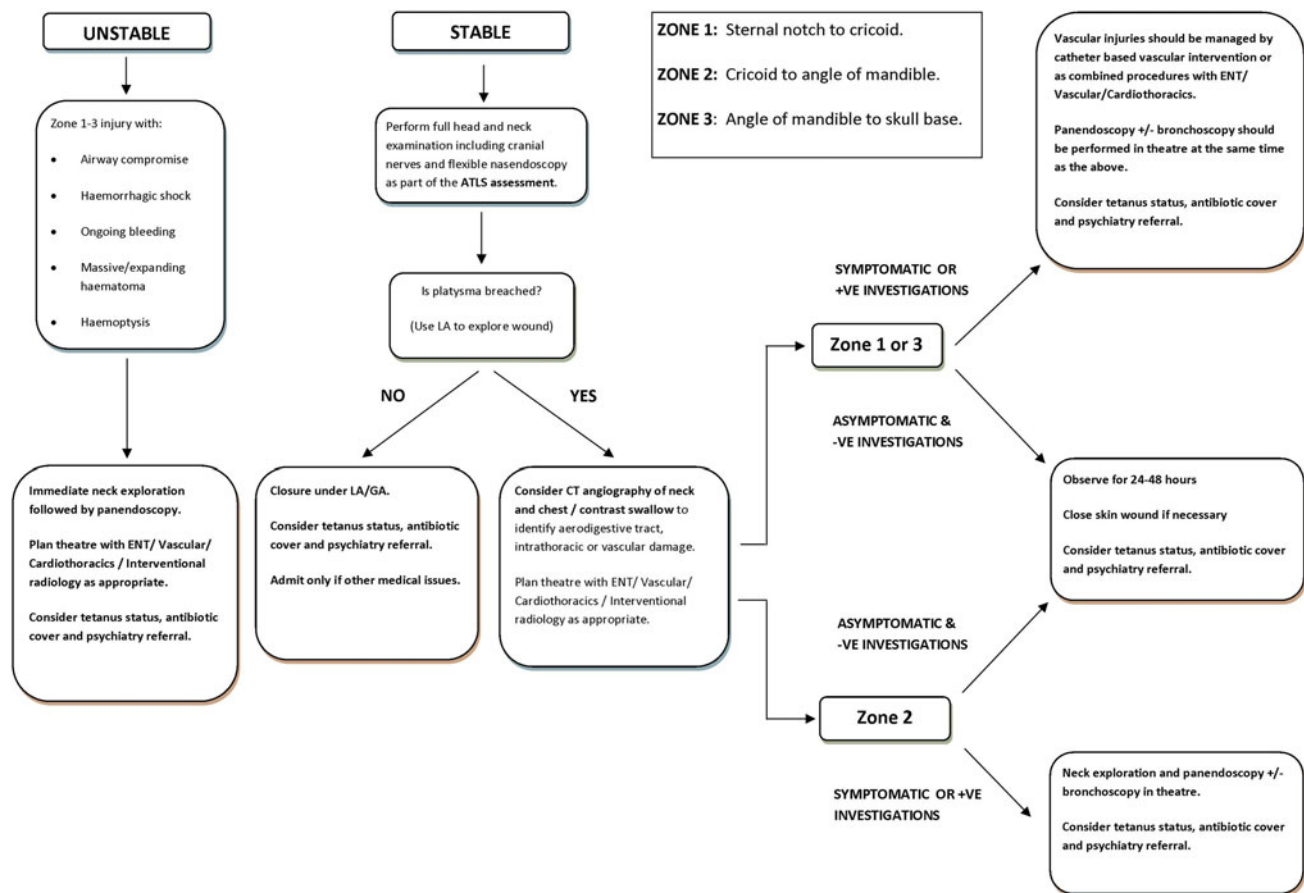


Fig. 1 The St George's Hospital guideline for the assessment and management of penetrating neck injuries

immediate exploration guided by clinical examination is both safe and cost-effective. The rationale for this is that vital structures in zone II are relatively accessible, in comparison to zone I and zone III. In addition, the sensitivity of physical examination for damage to vascular structures is validated for zone II trauma [10, 14, 30]. Damage to the upper aerodigestive tract occurs in over 20 % of PNIs [31]. From our data, one case (8.3 %) involved damage to the upper aerodigestive tract. Normal physical examination is not adequate to rule out upper aerodigestive tract damage [32, 33], whereas CT imaging can exclude these injuries in 90 % of cases [15]. It is important to rapidly identify and treat oesophageal damage in order to minimise patient morbidity following PNIs [34–36]. A contrast swallow examination in the acute setting may be of value [2, 37].

Whilst the evidence to rely on physical examination alone for zone II trauma is convincing, these recommendations generally reflect practice in high volume trauma centres in the United States. The volume of PNIs in the United Kingdom is relatively small and consequently trauma centres here are less experienced than their counterparts in other areas of the world at managing these injuries. Furthermore, dedicated head and

neck on-call emergency and out-of-hours cover is not universally present. For these reasons, the St George's algorithm suggests that the clinician considers CTA. CT imaging has the advantage of informing the surgeon about the presence of vascular, upper aerodigestive tract or intrathoracic injuries and allows them to plan their surgical approach accordingly. This may also include the involvement of another surgical discipline, for example cardiothoracics or vascular surgery, or interventional radiology. The use of CTA has also been shown to reduce the rate of negative surgical exploration [20–22]. It is suggested therefore, that the routine use of CT imaging in stable patients is safe and should form part of the management algorithm for penetrating neck injuries. It is stressed however that CTA should not be mandatory as a surgeon with the necessary expertise would be able to explore a zone II injury safely. This approach is increasingly supported in the literature [2, 37, 38].

Vascular injury

Damage to arteries following penetrating neck trauma may require repair by a vascular surgeon. Injuries to zone I requiring surgical exploration may require input from a

cardiothoracic surgeon, as treatment may require sternotomy or thoracotomy to gain proximal access to vascular structures. Similarly, arterial injuries within zone III or vertebral artery injuries are often best managed by catheter-based angiography and embolisation due to the difficulty in gaining surgical access [20]. Thus, the appropriate emergency cover specialists from vascular surgery, cardiothoracic surgery and interventional radiology should be consulted at an early stage to plan management accordingly. Out of the 12 injuries penetrating platysma in our series, four cases involved injury to zone I, none of which were discussed with cardiothoracics. One case involved a zone III injury, which required endovascular treatment by an interventional radiologist.

Conservative management

Although the management of PNI remains controversial, with no international consensus, there is a large volume of evidence that supports selective conservative management [8, 37, 38]. Patients who are ‘asymptomatic’ and have had negative investigations can be safely managed conservatively. For example, a patient who does not have hard signs of vascular or upper aerodigestive tract damage such as haematoma, ongoing haemorrhage, dysphonia, dysphagia and in whom a CTA was negative could be managed conservatively according to the St George’s guideline. This approach is similar to that described by Harris et al., who conclude that selective conservative management based on the absence of haemodynamic instability and radiological findings reduces length of hospital stay, lightens surgical workload and cuts costs without affecting morbidity or mortality [2]. It is recognised that these patients who are managed ‘conservatively’, i.e. do not have a formal neck exploration, may still require wound revision and skin closure.

The aetiology of PNIs in St George’s differs from that in the USA. The largest cause found in our series is deliberate self-harm, accounting for approximately half of the presentations. In contrast, the largest causes in the USA are gun shot wounds and knife stab wounds [8, 39]. The St George’s guideline acknowledges this difference by making reference to psychiatric referral where appropriate.

Conclusion

Despite being a UK trauma centre the volume of penetrating neck trauma presenting to St George’s Hospital is presently small compared to centres in the USA and South Africa. The presentations of injury are different with far fewer gunshot wounds and a greater proportion of deliberate self-harm. Because of the relative infrequency with

which penetrating neck injury presents to the on-call Otolaryngologist in the UK there is a need for penetrating neck injury guidelines. The difference in presentation of PNIs and the emergency cover structure in the UK means that guidelines from the USA or South Africa may not be directly applicable. The St George’s penetrating neck injury guideline has been developed with consideration of the pathology and services specific to this region of the UK and may be applicable to other UK centres. Clear guidelines should be drawn up for the management of PNIs in UK hospitals.

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Conflict of interest None.

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